

2017 Separable metric space.

✓ A metric space (X, d) is called separable if X has at least one countable dense subset i.e. $\exists$ a subset A of X such that

(i) A is countable
(ii) $\overline{A} = X$ i.e. A is dense in X

Ex^o Example:- $\mathbb{Q}$ is countable and $\overline{\mathbb{Q}} = \mathbb{R}$.

i.e. $\mathbb{R}$ is a separable metric space.
Similarly $\mathbb{R}^n$ is separable.

Theorem :-

✓ 2013 Totally bdd metric space is separable.

$\Rightarrow$ Let (X, d) be totally bdd metric space. Then for each $n=1, 2, 3, \dots$ and so on, $\exists$ an ε -nbd [finite number of open balls with radius ε], say E_n with $\varepsilon = \frac{1}{n}$.

Now $E = \bigcup E_n$ is ~~really~~ countable.

For any given $x \in X$, $d(x, E) \leq d(x, E_n) < \frac{1}{n} \quad \forall n$.

This implies $d(x, E) = 0$ as $n \rightarrow \infty$

$$\Rightarrow x \in \overline{E}$$

$$\Rightarrow \overline{E} = X$$

Thus X has a countable dense subset E . Hence X is separable.

Continuous funcⁿ on compact metric space

(i) $f: X \rightarrow Y$ is said to be continuous at $x_0 \in X$ iff for every sequence $\{x_n\}$ in X converging to x_0 , $\{f(x_n)\}$ converges to $f(x_0)$ in Y . i.e. $\lim_{n \rightarrow \infty} f(x_n) = f(\lim_{n \rightarrow \infty} x_n)$

Theorem :- $f: X \rightarrow Y$ is continuous on X iff for every open set (closed set) $G \subseteq Y$, $f^{-1}(G)$ is open (closed) in X .

Theorem (2014) Continuous image of a compact set is compact.

Proof:- Let $f: X \rightarrow Y$ be a continuous funcⁿ where X is compact and Y is any metric space. We will show that $f(X)$ is compact in Y .

Let $\{V_\alpha\}_{\alpha \in I}$ be an open cover of $f(X)$. Since f is continuous we know that each of the set $f^{-1}(V_\alpha)$ is open in X .

Since X is compact there are finitely many indices $\alpha_1, \alpha_2, \dots, \alpha_n$ such that $X \subseteq f^{-1}(V_{\alpha_1}) \cup f^{-1}(V_{\alpha_2}) \cup \dots \cup f^{-1}(V_{\alpha_n})$
 $\Rightarrow f(X) \subseteq V_{\alpha_1} \cup V_{\alpha_2} \cup \dots \cup V_{\alpha_n} [\because f(f^{-1}(E)) \subseteq E \text{ for every } E \subset Y]$

$\Rightarrow \{V_{\alpha_i}\}_{i=1}^n$ covers $f(X)$

Hence $f(X)$ is compact.

Prob:- Which of the following subset of $\mathbb{R}^2$ are compact

① $\{(x, y) : |x| \leq 1, |y| \geq 2\}$

② $\{(x, y) : |x| \leq 1, |y|^2 \leq 2\} \rightarrow \text{compact}$

③ $\{(x, y) : x^2 + 3xy^2 \leq 5\} \rightarrow \text{compact}$

④ $\{(x, y) : x^2 \leq y^2 + 5\}$



prob:- Let X be a metric space and $f: X \rightarrow \mathbb{R}$ be a cont. funcⁿ. Let $G = \{(x, f(x)) : x \in X\}$. Then G is homeomorphic to X .

(i) G is homeomorphic to $\mathbb{R}$

(ii) " " " " $X \times \mathbb{R}$

(iii) " " " " $\mathbb{R} \times X$

$$G = \{(x, f(x)) \mid x \in X\} \subseteq X \times \mathbb{R}$$

~~Prob~~ Uniform continuity on compact set :-

Continuous funcⁿ on a compact set is uniformly continuous

Ex: ~~2014~~ 2016
 (X, d) metric space, $d(x, A) = \inf\{d(x, y) \mid y \in A\}$
 $= f(x)$

$\therefore d(x, f(x))$ is u.c.

$\Rightarrow$ We have to prove that $|d(x, A) - d(z, A)| \leq d(x, z)$

let $\epsilon = \delta$. Then $d(x, z) < \delta$
 $\Rightarrow |f(x) - f(z)| < \epsilon \quad \forall x, z \in X$

let $y \in A$, then $d(x, y) \leq d(x, z) + d(z, y)$ [by triangle inequality]

Taking Infimum both side we get

$$d(x, A) \leq d(x, z) + d(z, A)$$

$$\Rightarrow |d(x, A) - d(z, A)| \leq d(x, z) < \delta = \epsilon$$

$\Rightarrow$ The funcⁿ is uniformly continuous.

prob:- $f: (0, 1) \rightarrow \mathbb{R}$

$f(x) = \frac{1}{x}, \sin \frac{1}{x}, \cos \frac{1}{x}$. is not u.c.

[Hints. f be continuous funcⁿ on $[a, b]$ and if the limit exist at the end point then f is u.c on $[a, b]$]

④ Prove that the funcⁿ $f: (0, 1] \rightarrow \mathbb{R}$ defined by $f(x) = \frac{1}{x}$ is not uniformly continuous.

$\Rightarrow f(x) = \frac{1}{x}$ is continuous on $(0, 1]$.

$$\lim_{x \rightarrow 0} f(x) = \infty$$

$\therefore$ The limit does not exist.

So by the statement 'f is continuous funcⁿ [a, b] and then f is u.c if the limit of f exists at the end points' the $f(x) = \frac{1}{x}$ is not uniformly continuous.

⑤ Prove that $\frac{\sin t}{t} \in [0, 1]$ is uniformly continuous on $(0, 1]$ but $\sin(\frac{1}{t})$ is not ^{uniformly} continuous on $[0, 1]$.

$$\Rightarrow f(x) = \frac{\sin t}{t}$$

$$\lim_{t \rightarrow 0} \frac{\sin t}{t} = \lim_{x \rightarrow 0} \left[\frac{0}{0} \right]$$

$$\lim_{t \rightarrow 0} \frac{\cos t}{1}$$

$$= 1$$

$\therefore f(x)$ is uniformly continuous on $(0, 1]$.

$$g(x) = \sin\left(\frac{1}{t}\right)$$

$$\lim_{t \rightarrow 0} g(x) = \infty$$

does not exist.

$\therefore f(x)$ is not uniform.